School of Information and Physical Sciences
SENG2250 System and Network Security
Assignment 3

Reflection

During the course of this project, I learned a lot about making web application with Flask, keeping passwords safe, using multi-factor authentication, and access control models. Putting together a safe web app with role-based access control taught me a lot of useful things. Knowledge from outside sources, like websites and documentation, as well as what I learned in class and in the lab were also important.

Learning from External Resources

To build this project, I relied heavily on various external sources to supplement what I learned in class. One example is Flask, a microweb platform that was the project's main part. The official Flask documentation (*Welcome to Flask — Flask Documentation, n.d)* was very helpful in showing me how to set up the app, make routes, handle requests, and write responses. With this basic understanding, I was able to add important features like JSON responses and authentication methods that use POST requests *(Innocent, 2024).*

The article on using Python to generate random passwords *(Bergantiños, 2024)* was another valuable resource that I used to help me create random passwords for new users. Furthermore, becoming knowledgeable with "bcrypt" hashing for safe password storing was essential for preventing security breaches on user credentials and integrating secure password verification into the login procedure (*How to Hash Passwords in Python?*, 2023).

Another condition of the project was to send MFA code by email. I used a guide on how to use the Mailgun API to put this together (*Learn How to Send Email | Mailgun*, 2024). Giving me the ability to send verification emails safely with multi-factor authentication (MFA) codes, which makes the system even safer.

Finally, I used Flask and JSON knowledge to make good use of my abilities to handle data. The JSON replies were easier to understand and safer when "jsonify" was used instead of "json.dumps" *(GeeksforGeeks, 2023).* This was especially important when working with private data because it made sure that the API replies were formatted correctly and had the right content type management.

Applying Concept from Labs and Lecture

Several of the main ideas I've learned in class are directly used in this project. These ideas are mostly about access control models and security. Implementing the Biba Access Control Model is one of the main ideas behind the system. This was talked about in class and practiced in lab. Users are limited by their security level and the security level of resources they access.

Furthermore, I applied the idea of least privilege, which ensures that users have access solely to the resources necessary for their roles. Admin users possess complete access to the system for user management, whilst standard users can only interact with data corresponding to their security clearance. This is implemented via the token-based authentication method.

Limitation and Solution

The program efficiently illustrates role-based access control and ensures authentication; however, it possesses certain limitations regarding scalability and security in practical applications.

Scalability: The current system can't grow very much because it stores data in flat files like users.txt. Since the number of users or data increases, it will take longer to access file, and problems may happen when more than one person tries to use the system at the same time. One way to solve this problem would be to use a more reliable database, like SQL or MongoDB, which works better for large-scale apps.

Security (Data transmission): Because the application implementation communicates via HTTP, private information like passwords and MFA codes is vulnerable to possible interception. By protecting the information sent between the client and the server, SSL/TLS encryption would solve this problem. It would guarantee that information, such as session tokens and user credentials, is encrypted while being transmitted, avoiding man-in-the-middle attacks.

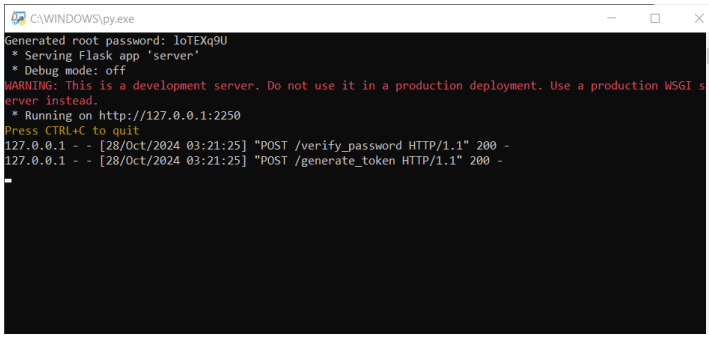
MFA and Email Security: Despite providing an additional degree of protection, the MFA system now uses email for verification, which isn't always the most practical or safest choice. Time-Based One-Time Password (TOTP) systems, such as Google Authenticator, would be an upgrade as they would generate tokens locally on the user's device, decreasing the need for email and enhancing security *(Multi-factor Authentication With TOTP (Google Authenticator) | Ory, 2024)*.

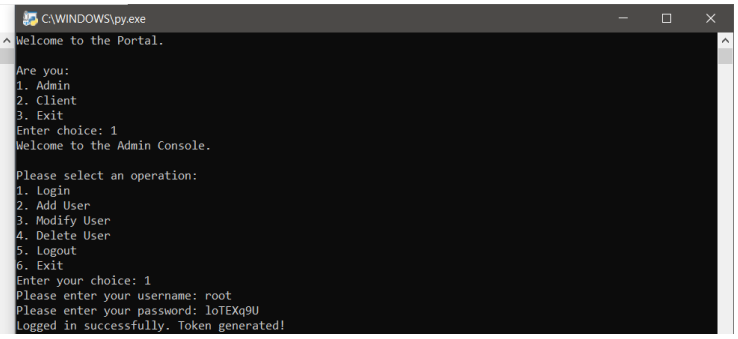
Conclusion

This project has given me the opportunity to practically apply important ideas from web development, security, and access control. Biba model implementation for data integrity, as well as learning how to safely handle user data and roles. My comprehension of the complexities involved in web application security has deepened. Scalability and security issues are among the system's drawbacks, but overall, it offers a strong platform for future advancement. The scalability and overall performance of the system might be greatly improved by putting technologies like databases and SSL/TLS into place.

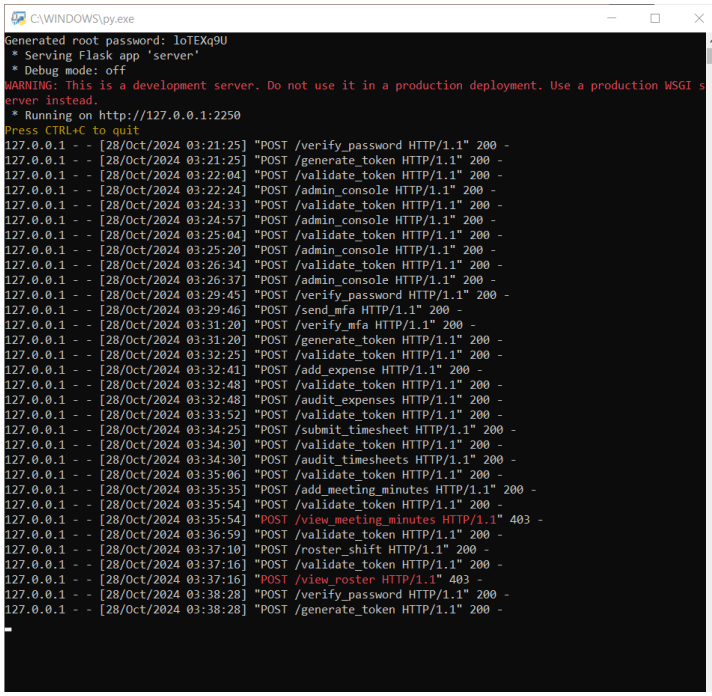
Test cases for system

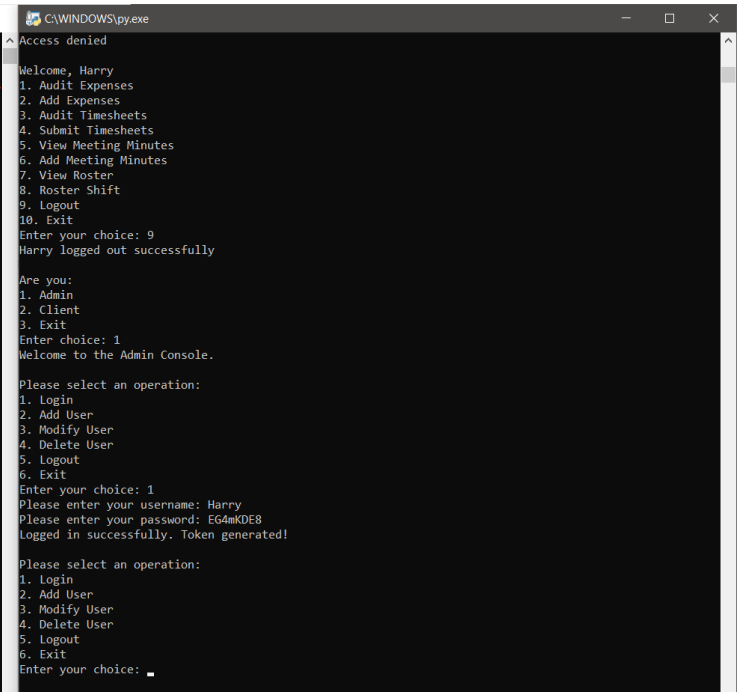
1. **Root user login**
    a. **Test root user password:** ensure root password is generated and printed on the server.
    Expected output: The root password should display on the server side when server initializes





2. **Admin group login**
    a. **Test admin login:** ensure user in admin group can login to admin console and use admin functionalities
    Expected output: Successfully login admin console and perform admin functions





```
C:\WINDOWS\py.exe
Generated root password: loTEXq9U
 * Serving Flask app 'server'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI s
erver instead.
 * Running on http://127.0.0.1:2250
Press CTRL+C to quit
127.0.0.1 - - [28/Oct/2024 03:21:25] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:21:25] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:22:04] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:22:24] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:24:33] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:24:57] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:25:04] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:25:20] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:26:34] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:26:37] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:29:45] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:29:46] "POST /send_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:31:20] "POST /verify_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:31:20] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:25] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:41] "POST /add_expense HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:48] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:48] "POST /audit_expenses HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:33:52] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:25] "POST /submit_timesheet HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:30] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:30] "POST /audit_timesheets HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:06] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:35] "POST /add_meeting_minutes HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:54] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:54] "POST /view_meeting_minutes HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:36:59] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:37:10] "POST /roster_shift HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:37:16] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:37:16] "POST /view_roster HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:38:28] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:38:28] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:39:40] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:39:46] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:39:56] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:39:57] "POST /send_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:11] "POST /verify_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:11] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:16] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:28] "POST /add_expense HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:40:32] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:32] "POST /audit_expenses HTTP/1.1" 200 -
```

```
C:\WINDOWS\py.exe
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 9
Harry logged out successfully

Are you:
1. Admin
2. Client
3. Exit
Enter choice: 1
Welcome to the Admin Console.

Please select an operation:
1. Login
2. Add User
3. Modify User
4. Delete User
5. Logout
6. Exit
Enter your choice: 1
Please enter your username: Harry
Please enter your password: EG4mKDE8
Logged in successfully. Token generated!

Please select an operation:
1. Login
2. Add User
3. Modify User
4. Delete User
5. Logout
6. Exit
Enter your choice: 3
Enter username to modify: Harry
Enter new group for Harry: admin
Enter new security level (TOP_SECRET / SECRET / UNCLASSIFIED): SECRET
User modified successfully!

Please select an operation:
1. Login
2. Add User
3. Modify User
4. Delete User
5. Logout
6. Exit
Enter your choice: 5
Harry logged out successfully!
```

b. **Test non-admin access admin's functions**: verify non-admin member cannot use admin functionalities

   Expected output: Receives a denied message when attempting to admin's functions (add, modify, delete user)

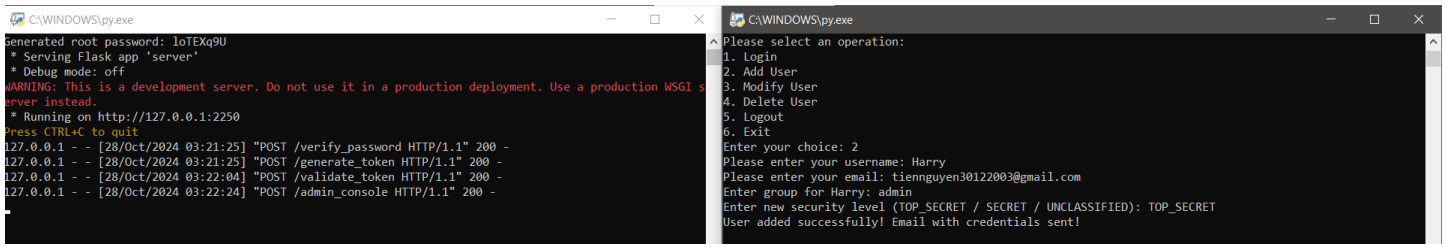
```
C:\WINDOWS\py.exe
27.0.0.1 - - [28/Oct/2024 03:51:54] "POST /view_roster HTTP/1.1" 200 -
27.0.0.1 - - [28/Oct/2024 04:04:06] "POST /validate_token HTTP/1.1" 401 -
27.0.0.1 - - [28/Oct/2024 04:07:25] "POST /verify_password HTTP/1.1" 200 -
27.0.0.1 - - [28/Oct/2024 04:07:25] "POST /generate_token HTTP/1.1" 200 -
27.0.0.1 - - [28/Oct/2024 04:07:27] "POST /validate_token HTTP/1.1" 200 -
27.0.0.1 - - [28/Oct/2024 04:07:36] "POST /admin_console HTTP/1.1" 200 -
27.0.0.1 - - [28/Oct/2024 04:07:43] "POST /verify_password HTTP/1.1" 200 -
27.0.0.1 - - [28/Oct/2024 04:07:43] "POST /generate_token HTTP/1.1" 200 -
27.0.0.1 - - [28/Oct/2024 04:08:55] "POST /verify_password HTTP/1.1" 200 -
27.0.0.1 - - [28/Oct/2024 04:08:55] "POST /generate_token HTTP/1.1" 200 -
27.0.0.1 - - [28/Oct/2024 04:10:18] "POST /validate_token HTTP/1.1" 200 -
27.0.0.1 - - [28/Oct/2024 04:10:26] "POST /admin_console HTTP/1.1" 403 -
```

```
C:\WINDOWS\py.exe
Please select an operation:
1. Login
2. Add User
3. Modify User
4. Delete User
5. Logout
6. Exit
Enter your choice: 1
Please enter your username: Harry
Please enter your password: EG4mKDE8
Logged in successfully. Token generated!

Please select an operation:
1. Login
2. Add User
3. Modify User
4. Delete User
5. Logout
6. Exit
Enter your choice: 2
Please enter your username: tien
Please enter your email: sdf
Enter group for tien: adf
Enter new security level (TOP_SECRET / SECRET / UNCLASSIFIED): SECRET
Access denied. Admins only!
```

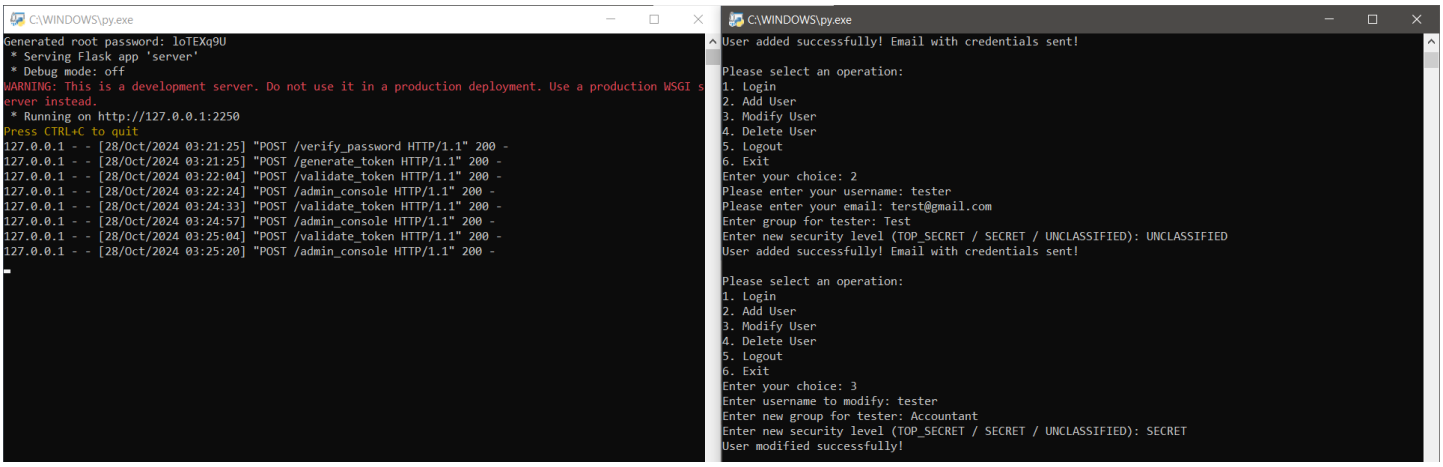
3. Admin functionalities

a. **Test adding a new user:** Verify an admin can add user to the system and specify user group and security level

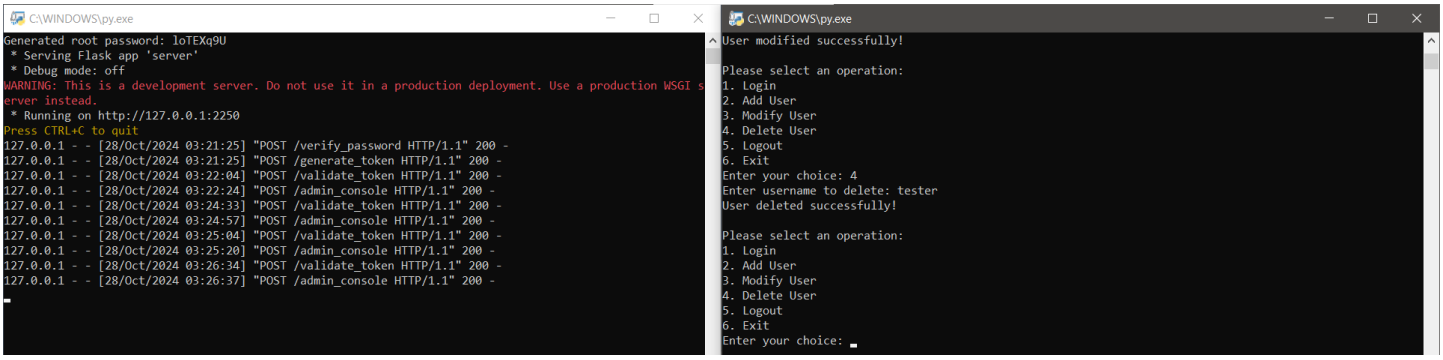
Expected output: New user data is correctly stored, and user receives and email with credentials.



b. **Test modifying user role and security level:** Ensure admin can change an existing user's group and security level
Expected output: The user's group and security level are updated, and the changes are reflected in subsequent access checks.

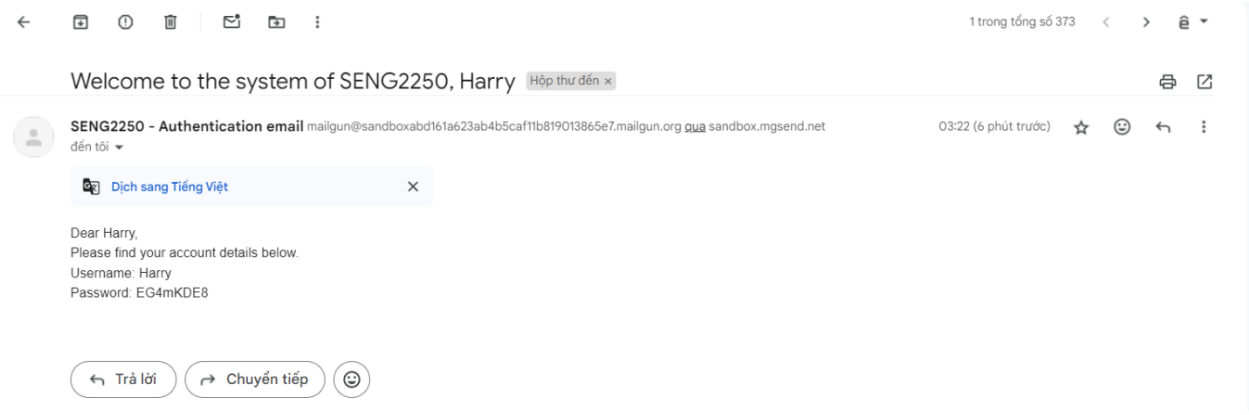


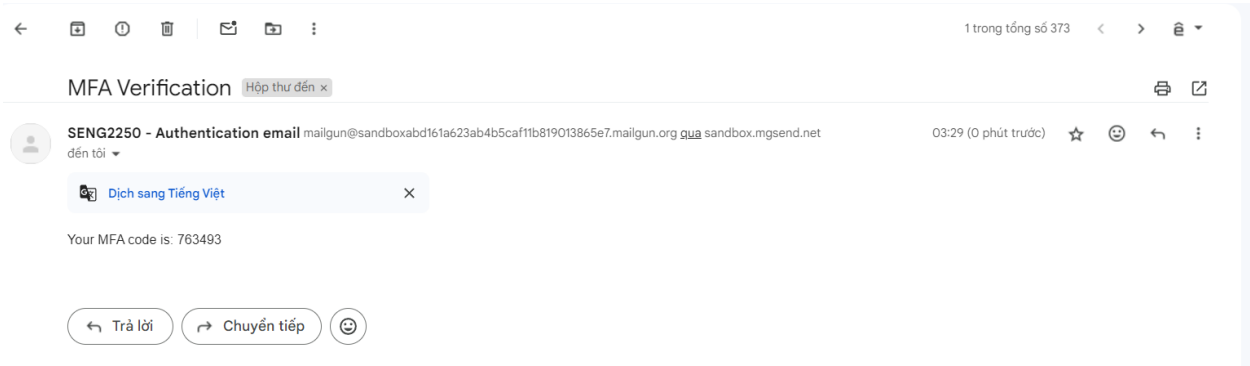
c. **Test deleting a user:** Verify admin can delete users
Expected output: User is removed from, and deleted user's access is denied immediately.

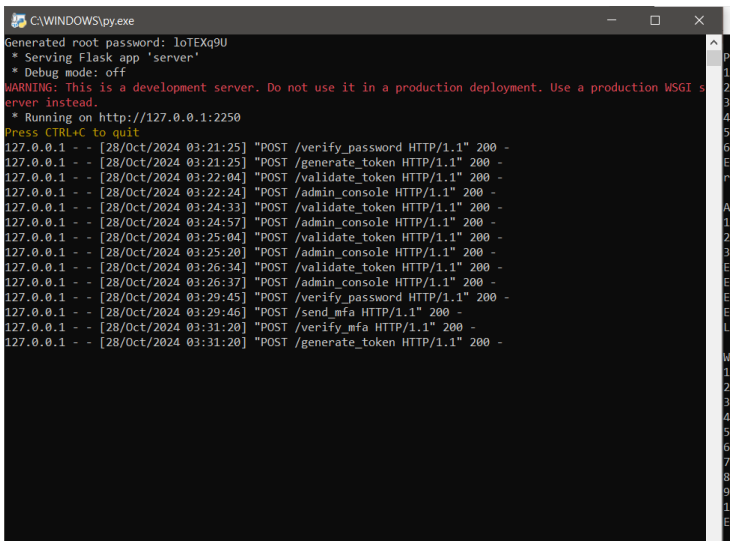


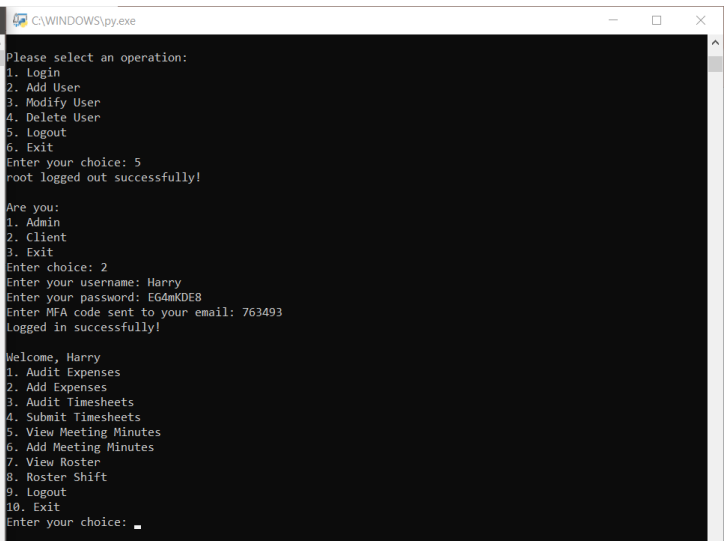
4. **User authentication**
   a. **Test user login with MFA verification:** Ensure that users must verify password then verify MFA code

Expected output: After successfully entered password, an MFA code is sent to user's email address. The user only login after entering correct MFA code.









b. **Test token expiry handling:** Verify token expiration after a set period (15 minutes)

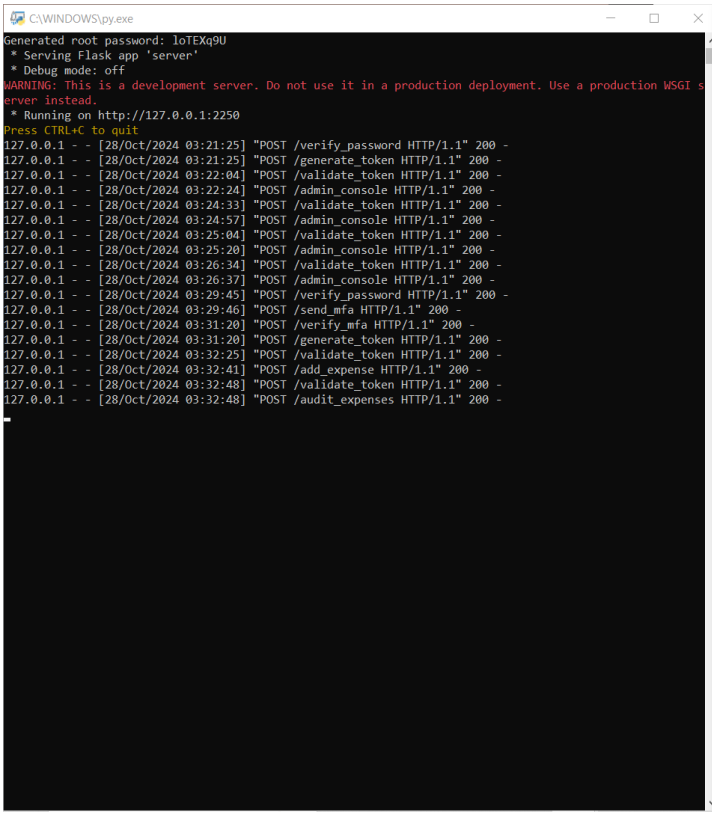
Expected output: If user tries to access a function after the token expiry time, the system should message them to log in again.

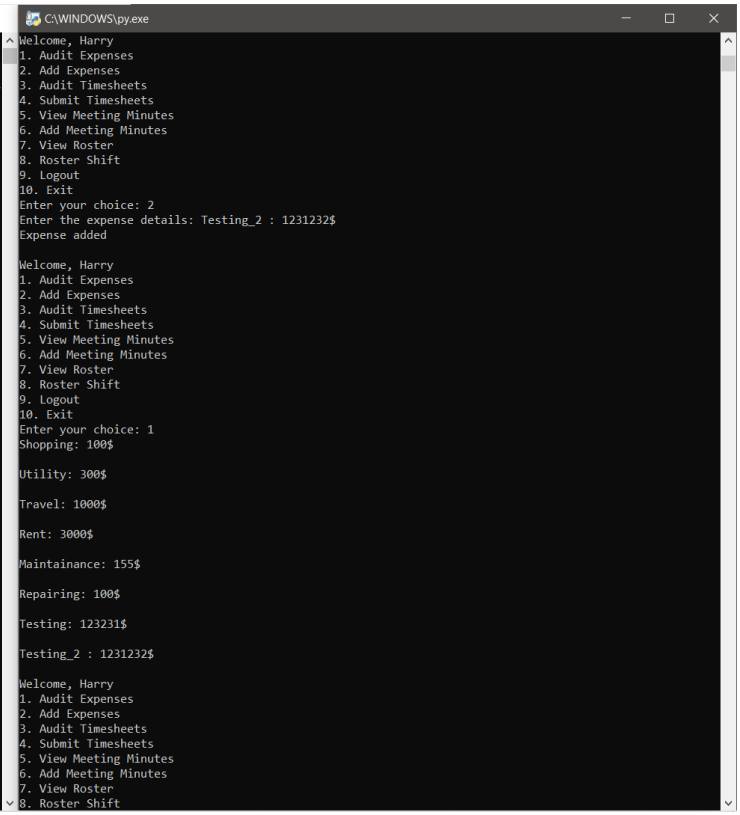
```
C:\WINDOWS\py.exe
127.0.0.1 - - [28/Oct/2024 03:40:32] "POST /audit_expenses HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:29] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:29] "POST /audit_timesheets HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:30] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:31] "POST /submit_timesheet HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:44:58] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:45:08] "POST /add_meeting_minutes HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:45:12] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:45:12] "POST /view_meeting_minutes HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:46:01] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:46:20] "POST /roster_shift HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:46:22] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:46:22] "POST /view_roster HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:47:14] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:47:14] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:47:16] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:47:28] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:48:43] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:48:44] "POST /send_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:48:57] "POST /verify_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:48:57] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:08] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:08] "POST /audit_expenses HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:09] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:12] "POST /add_expense HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:49:47] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:47] "POST /audit_timesheets HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:47] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:52] "POST /submit_timesheet HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:50:58] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:50:58] "POST /view_meeting_minutes HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:51:00] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:51:07] "POST /add_meeting_minutes HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:51:40] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:51:51] "POST /roster_shift HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:51:54] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:51:54] "POST /view_roster HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 04:04:06] "POST /validate_token HTTP/1.1" 401 -
```

```
C:\WINDOWS\py.exe
Enter your choice: 8
Enter the roster details: Troy(28/10): 10 hours
Shift rosted

Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 7
Troy(20/10): 5 hours

Harry(21/10): 8 hours

Miko(21/10): 3 hours

Linn(21/10): 7 hours
Harry(22/10): 3 hours
Harry(28/10): 12 hours
Jasmin(28/10): 8 hours
Troy(28/10): 10 hours

Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 3
Session expired! Please log in again to use those function.
```

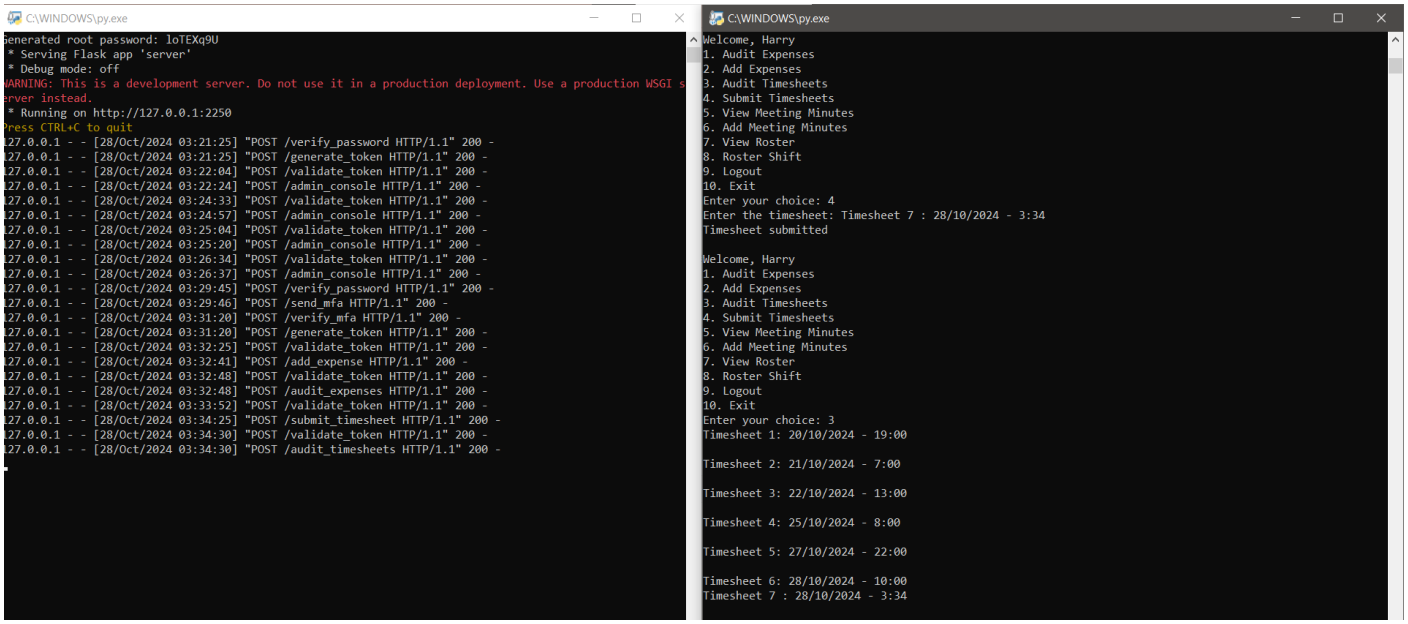
5. **Access control based on Security level (TOP_SECRET)**
    a. **Audit expenses and Add expenses:** Test if TOP_SECRET user can audit expense and add expenses

    Expected output: User can successfully add expenses and audit expenses





b. Audit timesheets and submit timesheet (same with audit and add expenses)

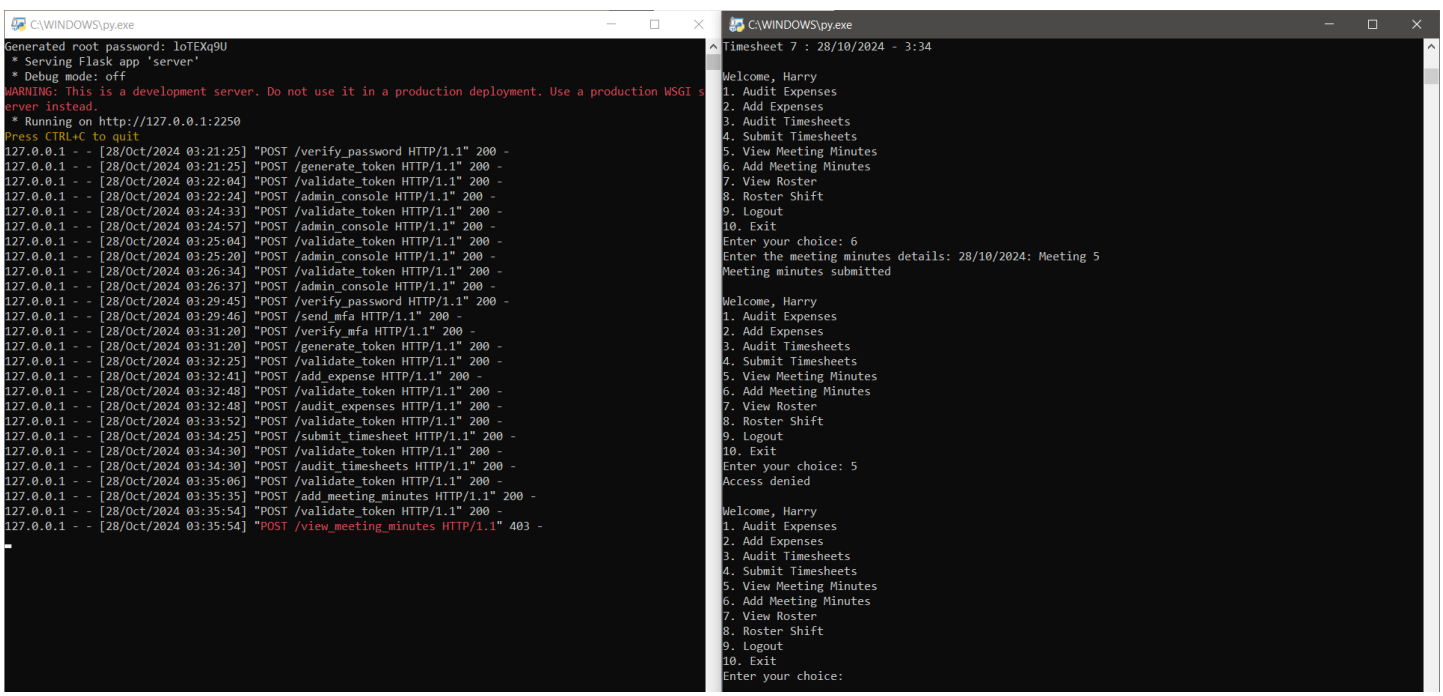


```
C:\WINDOWS\py.exe
Generated root password: loTEXq9U
 * Serving Flask app 'server'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI s
erver instead.
 * Running on http://127.0.0.1:2250
Press CTRL+C to quit
127.0.0.1 - - [28/Oct/2024 03:21:25] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:21:25] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:22:04] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:22:24] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:24:33] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:24:57] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:25:04] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:25:20] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:26:34] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:26:37] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:29:45] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:29:46] "POST /send_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:31:20] "POST /verify_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:31:20] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:25] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:41] "POST /add_expense HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:48] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:48] "POST /audit_expenses HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:33:52] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:25] "POST /submit_timesheet HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:30] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:30] "POST /audit_timesheets HTTP/1.1" 200 -
```

```
C:\WINDOWS\py.exe
Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 4
Enter the timesheet: Timesheet 7 : 28/10/2024 - 3:34
Timesheet submitted

Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 3
Timesheet 1: 20/10/2024 - 19:00

Timesheet 2: 21/10/2024 - 7:00

Timesheet 3: 22/10/2024 - 13:00

Timesheet 4: 25/10/2024 - 8:00

Timesheet 5: 27/10/2024 - 22:00

Timesheet 6: 28/10/2024 - 10:00
Timesheet 7 : 28/10/2024 - 3:34
```

c. View meeting minutes and add meeting minutes: Verify user can add meeting minutes but not view them (following Biba's model)

Expected output: User can add but receives an "Access Denied" response when attempting to view.



```
C:\WINDOWS\py.exe
Generated root password: loTEXq9U
 * Serving Flask app 'server'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI s
erver instead.
 * Running on http://127.0.0.1:2250
Press CTRL+C to quit
127.0.0.1 - - [28/Oct/2024 03:21:25] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:21:25] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:22:04] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:22:24] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:24:33] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:24:57] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:25:04] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:25:20] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:26:34] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:26:37] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:29:45] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:29:46] "POST /send_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:31:20] "POST /verify_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:31:20] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:25] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:41] "POST /add_expense HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:48] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:48] "POST /audit_expenses HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:33:52] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:25] "POST /submit_timesheet HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:30] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:30] "POST /audit_timesheets HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:06] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:35] "POST /add_meeting_minutes HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:54] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:54] "POST /view_meeting_minutes HTTP/1.1" 403 -
```

```
C:\WINDOWS\py.exe
Timesheet 7 : 28/10/2024 - 3:34

Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 6
Enter the meeting minutes details: 28/10/2024: Meeting 5
Meeting minutes submitted

Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 5
Access denied

Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice:
```

d. **View roster and roster shift:** Verify user can roster shifts but not view the roster.
Expected output: User can add a shift but denied access to view the roster.

```
C:\WINDOWS\py.exe
Generated root password: loTEXq9U
 * Serving Flask app 'server'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI s
erver instead.
 * Running on http://127.0.0.1:2250
Press CTRL+C to quit
127.0.0.1 - - [28/Oct/2024 03:21:25] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:21:25] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:22:04] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:22:24] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:24:33] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:24:57] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:25:04] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:25:20] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:26:34] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:26:37] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:29:45] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:29:46] "POST /send_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:31:20] "POST /verify_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:31:20] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:25] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:41] "POST /add_expense HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:48] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:48] "POST /audit_expenses HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:33:52] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:25] "POST /submit_timesheet HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:30] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:30] "POST /audit_timesheets HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:06] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:35] "POST /add_meeting_minutes HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:54] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:54] "POST /view_meeting_minutes HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:36:59] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:37:10] "POST /roster_shift HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:37:16] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:37:16] "POST /view_roster HTTP/1.1" 403 -
```

```
C:\WINDOWS\py.exe
Access denied

Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 8
Enter the roster details: Harry(28/10): 12 hours
Shift rosted

Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 7
Access denied

Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice:
```

6. **Access control based on Security level (SECRET)**
   a. **Audit expenses and Add expenses:** Verify that SECRET level user can audit expenses but cannot add expenses

      Expected output: User can view expenses, but it denied accessing when attempting to add them

```
C:\WINDOWS\py.exe
Generated root password: loTEXq9U
 * Serving Flask app 'server'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI s
erver instead.
 * Running on http://127.0.0.1:2250
Press CTRL+C to quit
127.0.0.1 - - [28/Oct/2024 03:21:25] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:21:25] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:22:04] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:22:24] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:24:33] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:24:57] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:25:04] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:25:20] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:26:34] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:26:37] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:29:45] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:29:46] "POST /send_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:31:20] "POST /verify_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:31:20] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:25] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:41] "POST /add_expense HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:48] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:32:48] "POST /audit_expenses HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:33:52] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:25] "POST /submit_timesheet HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:30] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:30] "POST /audit_timesheets HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:06] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:35] "POST /add_meeting_minutes HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:54] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:54] "POST /view_meeting_minutes HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:36:59] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:37:10] "POST /roster_shift HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:37:16] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:37:16] "POST /view_roster HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:38:28] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:38:28] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:39:40] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:39:46] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:39:56] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:39:57] "POST /send_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:11] "POST /verify_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:11] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:16] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:28] "POST /add_expense HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:40:32] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:32] "POST /audit_expenses HTTP/1.1" 200 -
```

```
Select C:\WINDOWS\py.exe
Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 2
Enter the expense details: Testing: 1200$
Access denied

Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 1
Shopping: 100$

Utility: 300$

Travel: 1000$

Rent: 3000$

Maintainance: 155$

Repairing: 100$

Testing: 123231$

Testing_2 : 1231232$

Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
```

b. Audit timesheets and submit timesheet (same with a)

```
C:\WINDOWS\py.exe
127.0.0.1 - - [28/Oct/2024 03:34:25] "POST /submit_timesheet HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:30] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:30] "POST /audit_timesheets HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:06] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:35] "POST /add_meeting_minutes HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:54] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:54] "POST /view_meeting_minutes HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:36:59] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:37:10] "POST /roster_shift HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:37:16] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:37:16] "POST /view_roster HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:38:28] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:38:28] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:39:40] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:39:46] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:39:56] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:39:57] "POST /send_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:11] "POST /verify_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:11] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:16] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:28] "POST /add_expense HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:40:32] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:32] "POST /audit_expenses HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:29] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:29] "POST /audit_timesheets HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:30] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:31] "POST /submit_timesheet HTTP/1.1" 403 -
```

```
C:\WINDOWS\py.exe
Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 3
Timesheet 1: 20/10/2024 - 19:00

Timesheet 2: 21/10/2024 - 7:00

Timesheet 3: 22/10/2024 - 13:00

Timesheet 4: 25/10/2024 - 8:00

Timesheet 5: 27/10/2024 - 22:00

Timesheet 6: 28/10/2024 - 10:00
Timesheet 7 : 28/10/2024 - 3:34

Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 4
Enter the timesheet: adsfsd
Access denied
```

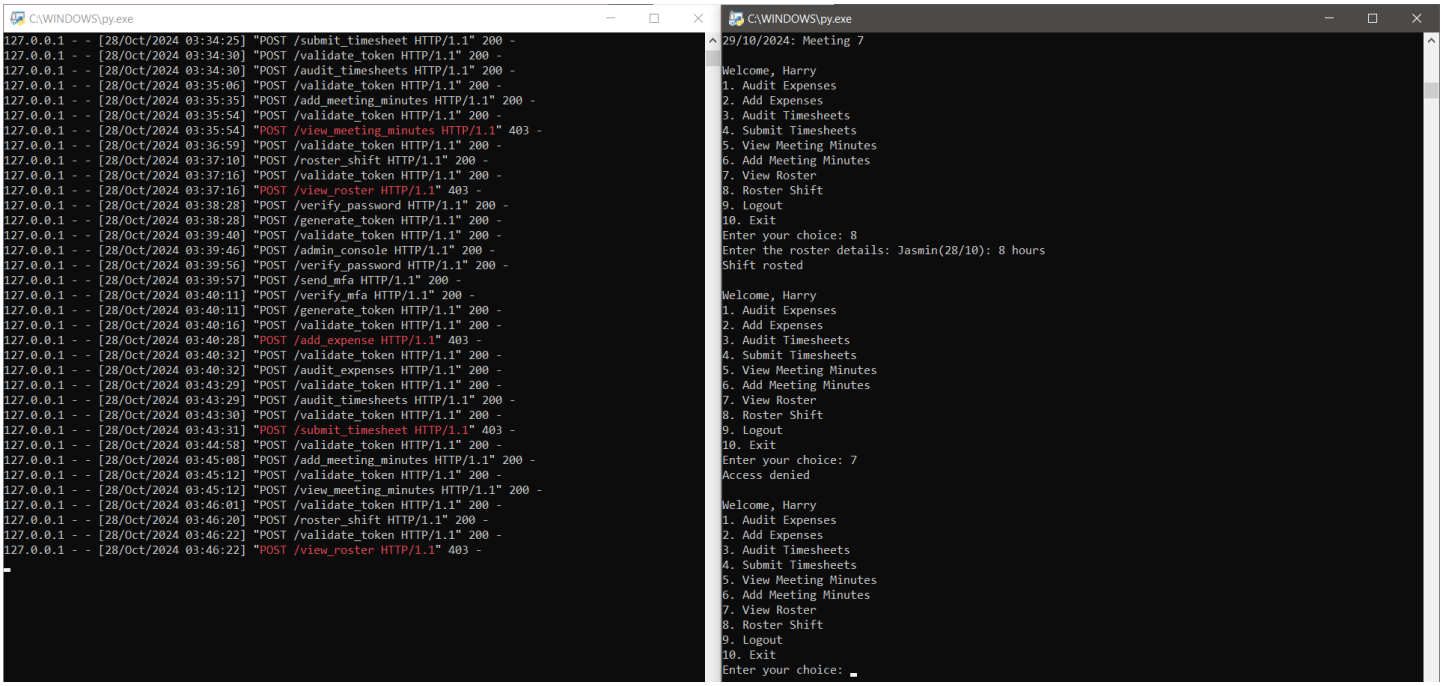
c. View meeting minutes and add meeting minutes: Ensure the user can access both viewing and adding meeting minutes

Expected output: User can successfully perform both actions

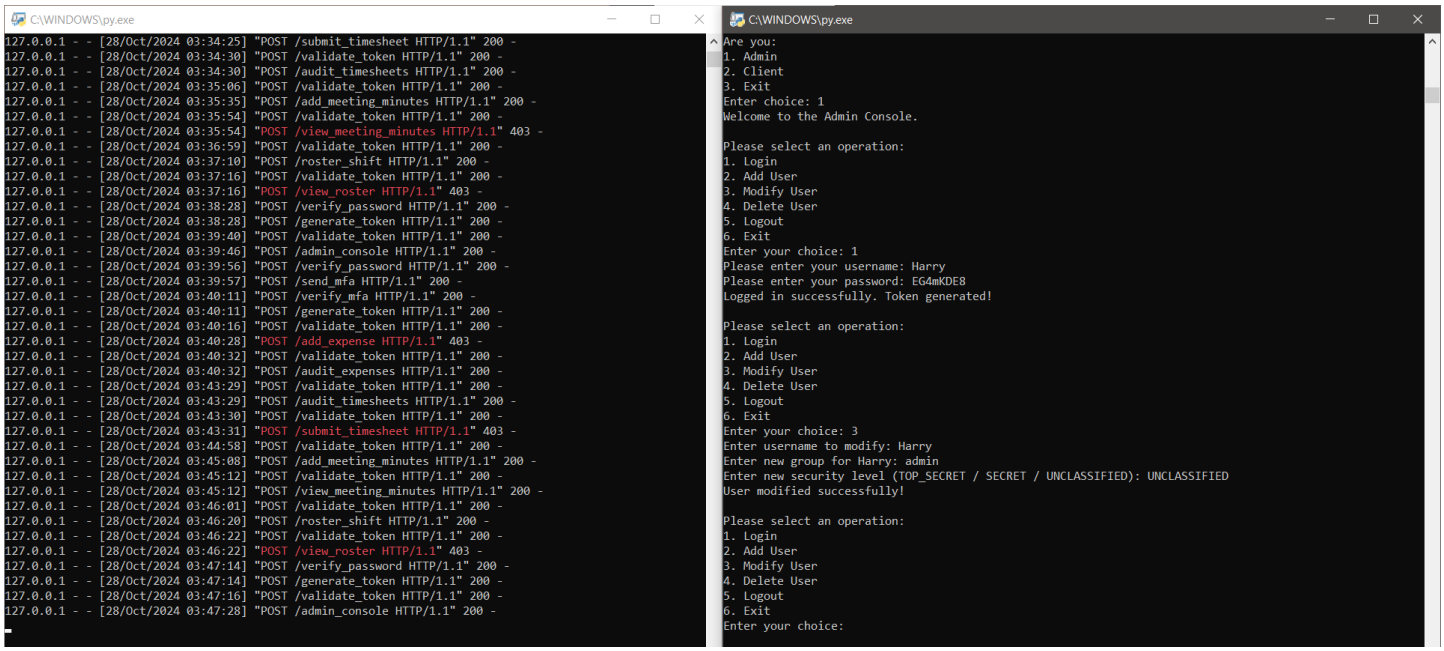
```
C:\WINDOWS\py.exe
127.0.0.1 - - [28/Oct/2024 03:34:25] "POST /submit_timesheet HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:30] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:34:30] "POST /audit_timesheets HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:06] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:35] "POST /add_meeting_minutes HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:54] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:35:54] "POST /view_meeting_minutes HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:36:59] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:37:10] "POST /roster_shift HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:37:16] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:37:16] "POST /view_roster HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:38:28] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:38:28] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:39:40] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:39:46] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:39:56] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:39:57] "POST /send_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:11] "POST /verify_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:11] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:16] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:28] "POST /add_expense HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:40:32] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:32] "POST /audit_expenses HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:29] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:29] "POST /audit_timesheets HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:30] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:31] "POST /submit_timesheet HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:44:58] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:45:08] "POST /add_meeting_minutes HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:45:12] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:45:12] "POST /view_meeting_minutes HTTP/1.1" 200 -
```

```
C:\WINDOWS\py.exe
Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 6
Enter the meeting minutes details: 29/10/2024: Meeting 7
Meeting minutes submitted

Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 5
15/10/2024: Kick-off meeting

20/10/2024: Meeting 1

25/10/2024: Meeting 2

26/10/2024: Meeting 3

27/10/2024: Meeting 4
28/10/2024: Meeting 5
```

d. **View roster and roster shift:** Confirm that the user can roster shift but cannot view the roster shift (UNCLASSIFIED)
Expected output: User successfully adds roster shift but receives an access error for viewing the roster.

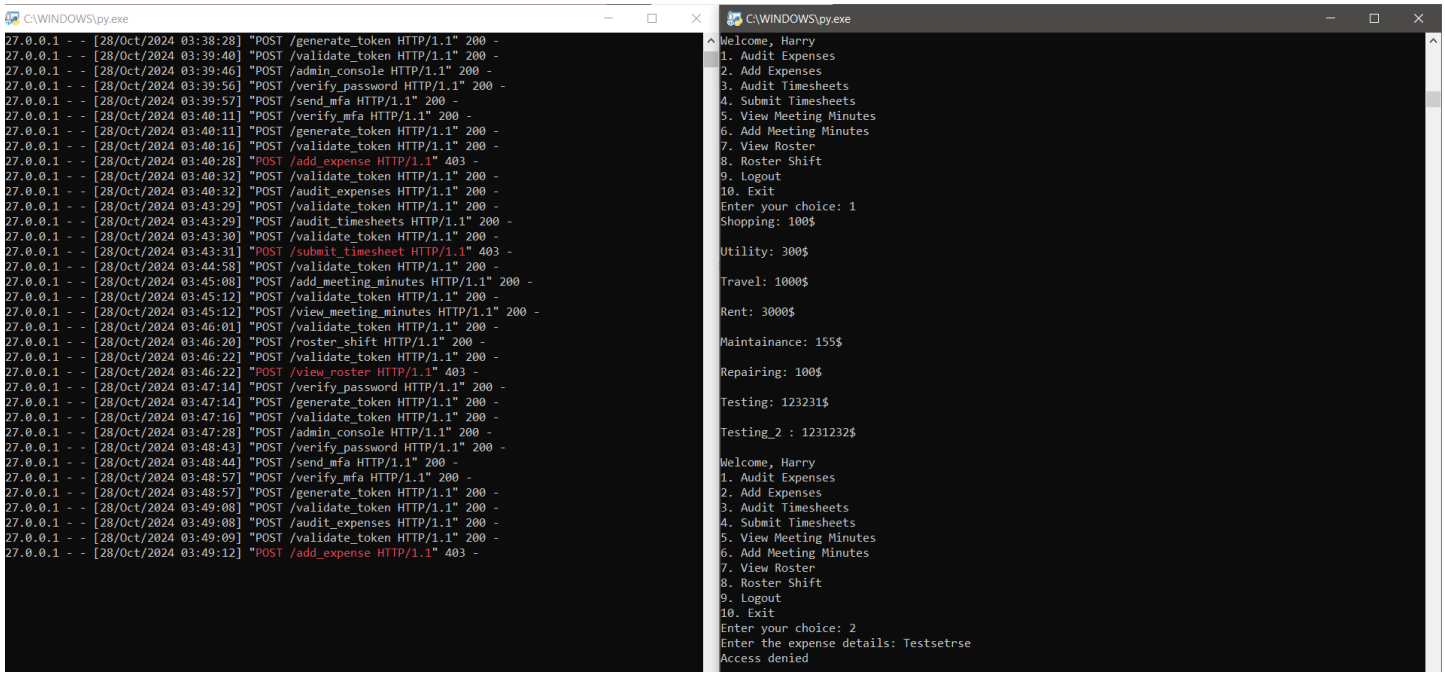


7. **Access control based on Security level (UNCLASSIFIED)**
Modify user security level to UNCLASSIFIED.



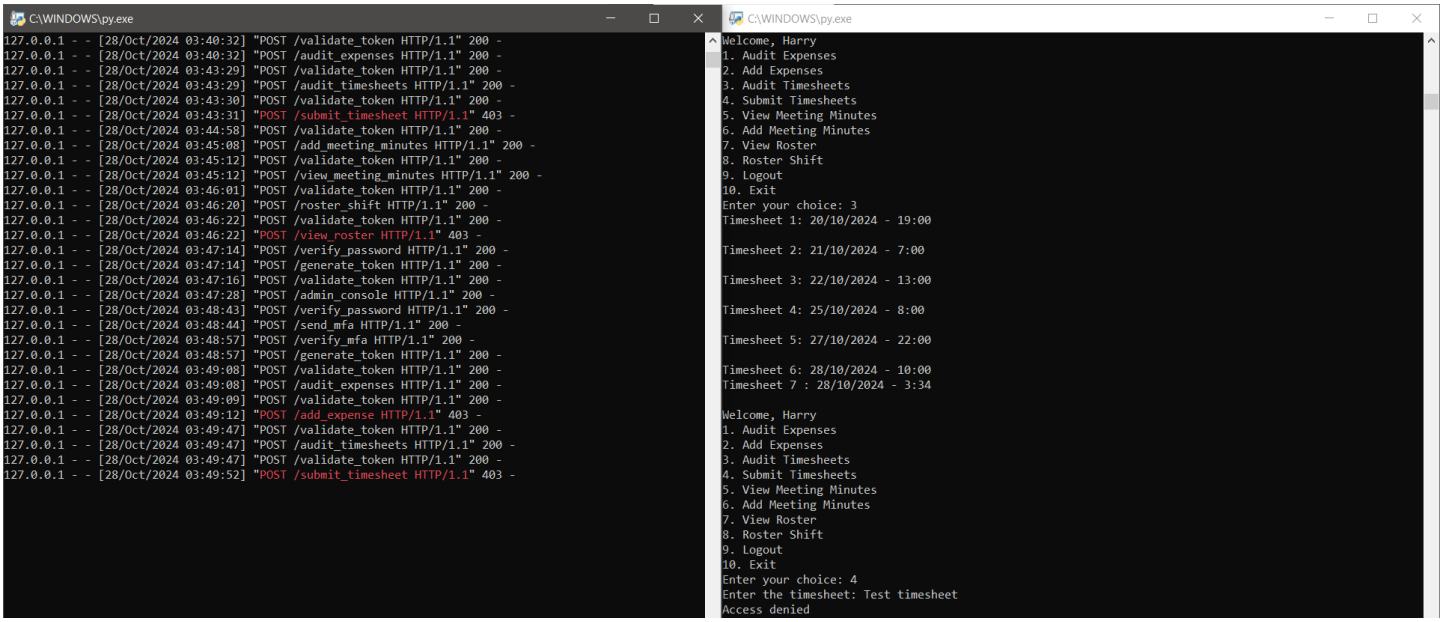
a. **Audit expenses and Add expenses:** Verify that an UNCLASSIFIED user can audit but cannot add expenses

Expected output: User can view expenses but denied when attempting to add expenses



b. **Audit timesheets and submit timesheet:**

Expected output: Audit succeeds but submitting results in an access error.



c. **View meeting minutes and add meeting minutes:** Test user can only view meeting minutes without the ability to add meeting minutes
Expected output: Viewing succeeds while adding meeting minutes receives an access denied.

```
C:\WINDOWS\py.exe
127.0.0.1 - - [28/Oct/2024 03:40:32] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:32] "POST /audit_expenses HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:29] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:29] "POST /audit_timesheets HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:30] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:31] "POST /submit_timesheet HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:44:58] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:45:08] "POST /add_meeting_minutes HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:45:12] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:45:12] "POST /view_meeting_minutes HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:46:01] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:46:20] "POST /roster_shift HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:46:22] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:46:22] "POST /view_roster HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:47:14] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:47:14] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:47:16] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:47:28] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:48:43] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:48:44] "POST /send_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:48:57] "POST /verify_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:48:57] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:08] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:08] "POST /audit_expenses HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:09] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:12] "POST /add_expense HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:49:47] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:47] "POST /audit_timesheets HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:47] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:52] "POST /submit_timesheet HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:50:58] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:50:58] "POST /view_meeting_minutes HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:51:00] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:51:07] "POST /add_meeting_minutes HTTP/1.1" 403 -
```

```
C:\WINDOWS\py.exe
Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 5
15/10/2024: Kick-off meeting

20/10/2024: Meeting 1

25/10/2024: Meeting 2

26/10/2024: Meeting 3

27/10/2024: Meeting 4
28/10/2024: Meeting 5
29/10/2024: Meeting 7

Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 6
Enter the meeting minutes details: meeting minutes test
Access denied
```

d. **View roster and roster shift:** Confirm the user can roster and view the roster shifts
Expected output: User can perform both functions

```
C:\WINDOWS\py.exe
127.0.0.1 - - [28/Oct/2024 03:40:32] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:40:32] "POST /audit_expenses HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:29] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:29] "POST /audit_timesheets HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:30] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:43:31] "POST /submit_timesheet HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:44:58] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:45:08] "POST /add_meeting_minutes HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:45:12] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:45:12] "POST /view_meeting_minutes HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:46:01] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:46:20] "POST /roster_shift HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:46:22] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:46:22] "POST /view_roster HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:47:14] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:47:14] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:47:16] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:47:28] "POST /admin_console HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:48:43] "POST /verify_password HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:48:44] "POST /send_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:48:57] "POST /verify_mfa HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:48:57] "POST /generate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:08] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:08] "POST /audit_expenses HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:09] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:12] "POST /add_expense HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:49:47] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:47] "POST /audit_timesheets HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:47] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:49:52] "POST /submit_timesheet HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:50:58] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:50:58] "POST /view_meeting_minutes HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:51:00] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:51:07] "POST /add_meeting_minutes HTTP/1.1" 403 -
127.0.0.1 - - [28/Oct/2024 03:51:40] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:51:51] "POST /roster_shift HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:51:54] "POST /validate_token HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2024 03:51:54] "POST /view_roster HTTP/1.1" 200 -
```

```
C:\WINDOWS\py.exe
Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 8
Enter the roster details: Troy(28/10): 10 hours
Shift rosted

Welcome, Harry
1. Audit Expenses
2. Add Expenses
3. Audit Timesheets
4. Submit Timesheets
5. View Meeting Minutes
6. Add Meeting Minutes
7. View Roster
8. Roster Shift
9. Logout
10. Exit
Enter your choice: 7
Troy(20/10): 5 hours

Harry(21/10): 8 hours

Miko(21/10): 3 hours

Linn(21/10): 7 hours
Harry(22/10): 3 hours
Harry(28/10): 12 hours
Jasmin(28/10): 8 hours
Troy(28/10): 10 hours
```

Reference

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